Autonomous Drone Simulation

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Project Scope

- Create a drone show simulator to control several drones. You should be able to make a mission plan for the drones to create a custom drone show. Drones should not collide and should move to the next shape efficiently.

Objectives

Presthika:

- Implement the class for Drone. This will have attributes like the position, velocity, battery level, and mission status. There will also be the methods and status updates.
- Develop the collision detection and avoidance modules.
- Write tests to test and validate the drone and edge cases.
- Learning more about managing a team.
- Learning more in depth concepts of C++
- Learn how to organize and have a put together GitHub.

Riley:

- Implement FleetMananger class. This will coordinate the drones and their missions.
- Find the most optimal path for each drone to go to their next position
- Develop the mission planning algorithm
- Build simulation loop and the display logic

Learning objectives:

- Learn more about class structures and using them in a large scale
- Learn about graphics and how to make a GUI in C++

Team Goals/Objectives:

- Build this autonomous drone simulation with coordination between at least 3 drones
- Collision avoidance strategy
- Make sure everything is well documented and works
- Allow user inputs and variations to create different shapes
- Learn how to manage a large scale OOP project
- Learn how to manage a GitHub project
- Learn how to manage a team during a team project

Technology and Tools

- C++
- VSCODE
- GitHub
- Copilot

Timeline:

- Starting from now (counting of weeks)
- Week 1: Finalize the plan, agree on tasks, set up Git and make sure everyone has access to all the files needed.
- Week 2: Build the Drone class, FleetManager Class, and the simulation loop.
- Week 3: Integrate everything and then implement the basic missions
- Week 4: Add in collision detection, continue testing and making edits for different bugs and cleaning up the code as needed.
- Week 5: Run simulations and fix as needed. Finalize and push to Git and keep testing and changing if needed.

Github Repo Link: https://github.com/rileyashok/DroneSims

Project Timeline:
https://northeastern-my.sharepoint.com/:x:/r/personal/vijaykumar_pr_northeastern_edu/_layouts/15/Doc.aspx?sourcedoc=%7B293D77D0-D34E-44F7-B46B-67FF811E3CEF%7D&file=Project_Schedule_Presthika%20and%20Riley.xlsx&fromShare=true&action=default&mobileredirect=true

PROJECT SCHEDULE TEMPLATE							
PROJECT NAME	PROJECT MANAGER						
PROJECT DELIVERABLE							
SCOPE STATEMENT							
START DATE	END DATE	OVERALL PROGRESS					
2/25/2026	4/8/2026	5%					
TASK NAME	ASSIGNED TO	START DATE	END DATE	DURATION (in days)	STATUS	COMMENTS	Not Started
Finalize Plan	Both	25-Feb	27-Feb	2	Complete		Complete
Build Drone class	Riley	27-Feb	13-Mar	14	Not Started		Overdue
Fleet Manager Class	Presthika	27-Feb	13-Mar	14	Not Started		In Progress
Simulation loop	Riley	11-Mar	3/20/2026	9	Not Started		
User interface/graphics	Presthika	3/20/2026	3/27/2026	7	Not Started		
Collision Detection	Riley	2/28/2026	3/13/2026	13	Not Started		
Testing	Both	3/27/2026	4/3/2026	7	Not Started		
Presentation	Both	4/3/2026	4/10/2026	7	Not Started		
Technical Report	Both	4/3/2026	4/10/2026	7	Not Started		
				0	Not Started		
				0	Not Started		
				0	Not Started		
				0	Not Started		
				0	Not Started		